What is the most probable distance of a 1s electron in a He^+ ion. The wave function for 1s orbital is given by $\Psi = \sqrt{\left(\frac{Z^3}{\pi a_o^3}\right)} e^{-zr/a_o}$, where a_o = radius of first Bohr's orbit in H-atom = 52.9 pm.

(A) 52.9 pm (B) 13.25 pm (C) 6.61 pm (D) 26.45 pm

Imp

He^+, 1s : $1s^1$ (single e^- species)
$(Z=2)$

$\psi_{1s} = \sqrt{\frac{Z^3}{\pi a_0^3}} \times e^{-Zr/a_0}$.

Radial distribution function $= 4\pi r^2 \psi_r^2$

$= 4\pi r^2 \times \left(\sqrt{\frac{Z^3}{\pi a_0^3}}\right)^2 \times e^{-\frac{2Zr}{a_0}}$

→ For maximum probability of finding e^- in 1s, $\frac{d}{dr}(4\pi r^2 \psi_r^2) = 0$

$\frac{d}{dr}\left\{\left(4\pi \times \frac{Z^3}{\pi a_0^3}\right) \times r^2 e^{-\frac{2Zr}{a_0}}\right\} = 0$

$\cancel{r} \times \left(\cancel{e^{-\frac{2Zr}{a_0}}}\right) \times \left(\frac{-2Z}{a_0}\right) + \left(\cancel{e^{-\frac{2Zr}{a_0}}}\right) \times 2\cancel{r} = 0$

$-\frac{\cancel{2}Z}{a_0} \times r_{max} + \cancel{2} = 0$

$r_{max} = \frac{a_0}{2}$

$= 52.9/2$

$= 26.45$ pm

The maximum number of electrons that can have principal quantum number, n = 3, and spin quantum number, $m_s = -1/2$, is

Ans : 9

Sol^n: $n=3$, $s=-1/2$ ($\underbrace{s \text{ or } m_s}$ → spin quantum Number | (m or m_l) → magnetic quantum No.

3s, 3p, 3d.

$\underbrace{\boxed{\uparrow}, \boxed{\uparrow|\uparrow|\uparrow} \boxed{\uparrow|\uparrow|\uparrow|\uparrow|\uparrow}}$

Ans : 9

9 e^- having $n=3$ & $s=-1/2$

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Not considering the electronic spin, the degeneracy of the second excited state (n = 3) of H atom is 9, while the degeneracy of the second excited state of H^- is

Ans: 3

Sol^n: degenerate orbitals: orbitals having equal energy.

1 3 5 orbitals.

H: $1s^1$, $\underbrace{2s = 2p}_{}$, $\underbrace{3s = 3p = 3d}_{}$

(single e^- species)

equal energy — 1st excited state

equal energy. degeneracy = 9

H^-: $1s^2$, $2s$, $2p$ → 3 degenerate orbitals

(Multi e^- species) (Total $2e^-$)

1st excited state

2nd excited state.

$\underbrace{(2p_x, 2p_y, 2p_z)}_{3}$

(n+l) rule

Ans: 3

atoms

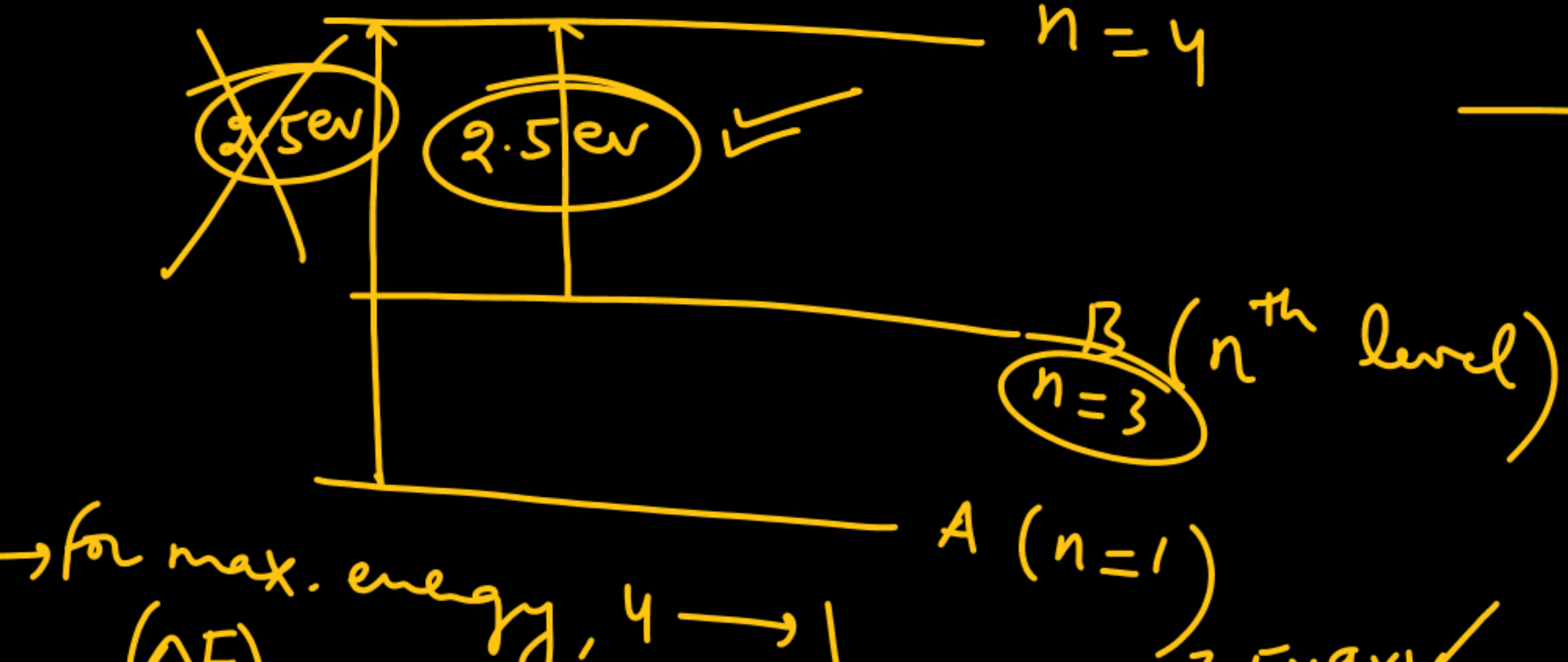
A gas of identical H-like atom has some atoms in the lowest (ground) energy level A and some stoms in a particular upper (excited) energy level B and there are no atoms in any other energy level. The atoms of the gas make transition to a higher energy level by absorbing monochromatic light of photon energy 2.5 eV. Subsequently, the atoms emit radiation of only six different photons energies. Some of the emitted photons have energy 2.5 eV. Some have more than 2.5 eV energy and no photon has less than 2.5 eV energy. Find the maximum energy of the emitted photon.

Ans : 48.2 eV.

$$\begin{array}{r} 135 \\ \times\ 2.5 \\ \hline 675 \\ 270 \\ \hline 337.5 \end{array}$$

$\frac{337.5}{7} = 48.2\ eV$

Sol^n:



→ No. of transitions or photons emitted $= \frac{n(n-1)}{2} = 6$

$n = 4$

→ Energy 2.5 eV is absorbed by atoms in excited state (B)

$$KZ^2\left(\frac{1}{3^2} - \frac{1}{4^2}\right) = 2.5$$

$$KZ^2 \times \frac{7}{9 \times 16} = 2.5 \Rightarrow KZ^2 = \frac{2.5 \times 9 \times 16}{7}$$

→ for max. energy, $4 \rightarrow 1$

$$(\Delta E)_{max} = E_4 - E_1 = KZ^2\left(\frac{1}{1^2} - \frac{1}{4^2}\right) = \frac{2.5 \times 9 \times \cancel{16}}{7} \times \frac{15}{\cancel{16}} = \frac{2.5 \times 135}{7}$$

Photochemical dissociation of oxygen result in the production of two oxygen atoms, one in the ground state and one in the excited state [O*].

$$O_2 \xrightarrow{h\nu} O + O^*$$

The maximum wavelength (λ) needed for this is 160 nm. If the excitation energy $O \longrightarrow O^*$ is 4×10^{-20} J. How much energy in eV is needed for the dissociation of one oxygen molecule into normal atoms in ground state ? (Given hc = 12400 eV Å)

Ans : 7.5 eV

in eV

$$E = \frac{12400}{\lambda (\text{in Å})}$$

Sol^n :

$$O_2 \xrightarrow{x\ eV} O + O$$

$O \xrightarrow{0} O$, $O \xrightarrow{4\times10^{-20}\ \text{Joule}} O^*$

$O_2 \xrightarrow{160\ nm} O + O^*$

$$\frac{\cancel{12400}^{31}}{\cancel{1600}_{4}} eV = x + 0 + \frac{\cancel{4}\times 10^{-20}}{\cancel{1.6}\times 10^{-19}} eV.$$

$$7.75 = x + \cancel{0.25}\ 0.4$$

$$\Rightarrow \boxed{x = 7.5 eV}\ \text{Ans.}$$

$1\ eV = 1.6 \times 10^{-19}$ J.

$1\ J = \frac{1}{1.6\times10^{-19}}$ eV.

1 eV/atom $\cong$ 96.4 KJ/mol

$\cong \frac{96.4}{4.18}$ KCal/mol

$\cong$ 23 KCal/mol

Infrared lamps are used in restaurants to keep the food warm. The infrared radiation is strongly absorbed by water, raising its temperature and that of the food. If the wavelength of infrared radiation is assumed to be 1500 nm, then the number of photons per second of infrared radiation produced by an infrared lamp that consumes energy at the rate of 100 W and is 12% efficient only is ($x \times 10^{19}$). The value of x is; (Given : $h = 6.625 \times 10^{-34}$ J–s)

Solⁿ : $\lambda = 1500$ nm, ~~x photons/sec~~ . $x \times 10^{19}$ photons/sec.

$P = 100W$, $\eta = 12\% = 0.12$

100×0.12 Joule $= x \times 10^{19} \times \frac{hc}{\lambda}$

$$12 = x \times 10^{19} \times \frac{6.625 \times 10^{-34} \times 3 \times 10^{8}}{1500 \times 10^{-9}}$$

$$12 = x \times \cancel{10^{-6}} \times \frac{6.625}{15\cancel{00}} \times 3 \times \cancel{10^{8}}$$

$$12 = x \times \frac{6.625}{\cancel{15}\,5} \times \cancel{3}$$

$$x = \frac{60}{6.625}$$

$\simeq 9$ Ans.

If λ_0 is the threshold wavelength for photoelectric emission from a metal surface, λ is the wavelength of light falling on the surface of metal and m is the mass of electron, then the maximum speed of ejected electrons is given by

(A) $\left[\frac{2h}{m}(\lambda_0 - \lambda)\right]^{1/2}$ (B) $\left[\frac{2hc}{m}(\lambda_0 - \lambda)\right]^{1/2}$ (C) $\left[\frac{2hc}{m}\left(\frac{\lambda_0 - \lambda}{\lambda_0 \lambda}\right)\right]^{1/2}$ (D) $\left[\frac{2h}{m}\left(\frac{1}{\lambda_0} - \frac{1}{\lambda}\right)\right]^{1/2}$

Sol^n: λ should be less than λ_0.

$$(KE)_{\substack{\text{photoelectron} \\ \text{max}}} = \frac{hc}{\lambda} - \frac{hc}{\lambda_0}$$

$$\frac{1}{2} m \times v_{max}^2 = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

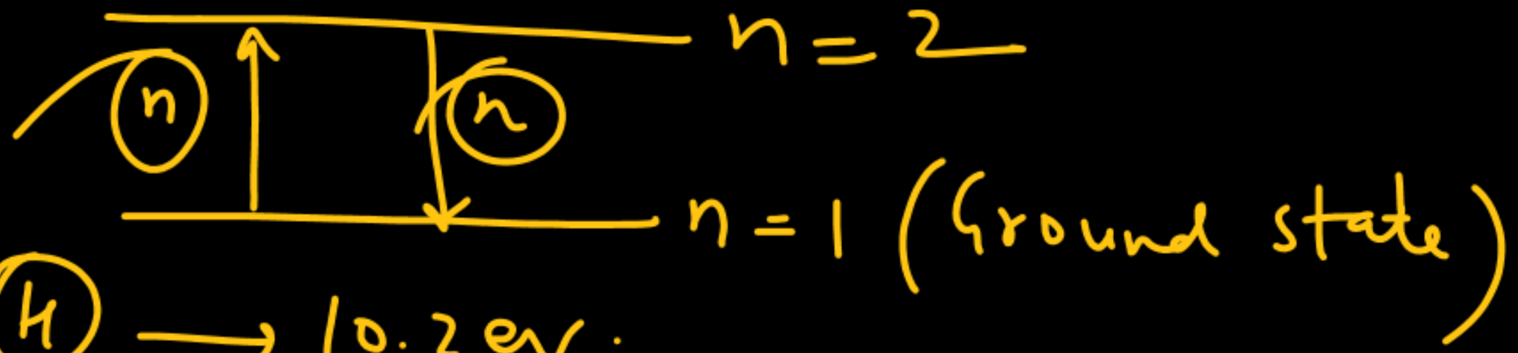
$$= hc\left(\frac{\lambda_0 - \lambda}{\lambda \lambda_0}\right)$$

$$v_{max} = \sqrt{\frac{2hc}{m}\left(\frac{\lambda_0 - \lambda}{\lambda \lambda_0}\right)}$$

Some hydrogen-like atoms in ground state absorbs ‘n’ photons having same the energy and on de-excitement, it emits exactly ‘n’ photons. The energy of absorbed photon may be

(A) 91.8 eV (B) 40.8 eV (C) 48.4eV (D) 54.4 eV

Sol^n :



H $\longrightarrow$ 10.2 eV.

$He^{+} \longrightarrow 2^2 \times 10.2 = 40.8$ eV

$Li^{2+} \longrightarrow 3^2 \times 10.2 = 91.8$ eV

$Be^{3+} \longrightarrow 4^2 \times 10.2 = 163.2$ eV

The correct statement(s) regarding $3p_y$ orbital is/are

(A) angular part of wave function is independent of angles θ and ϕ.

(B) number of maxima in $4\pi r^2 R^2(r)$ vs. r curve is 2.

(C) XZ plane is the nodal plane.

(D) magnetic quantum number must be – 1.

Ans : B, C

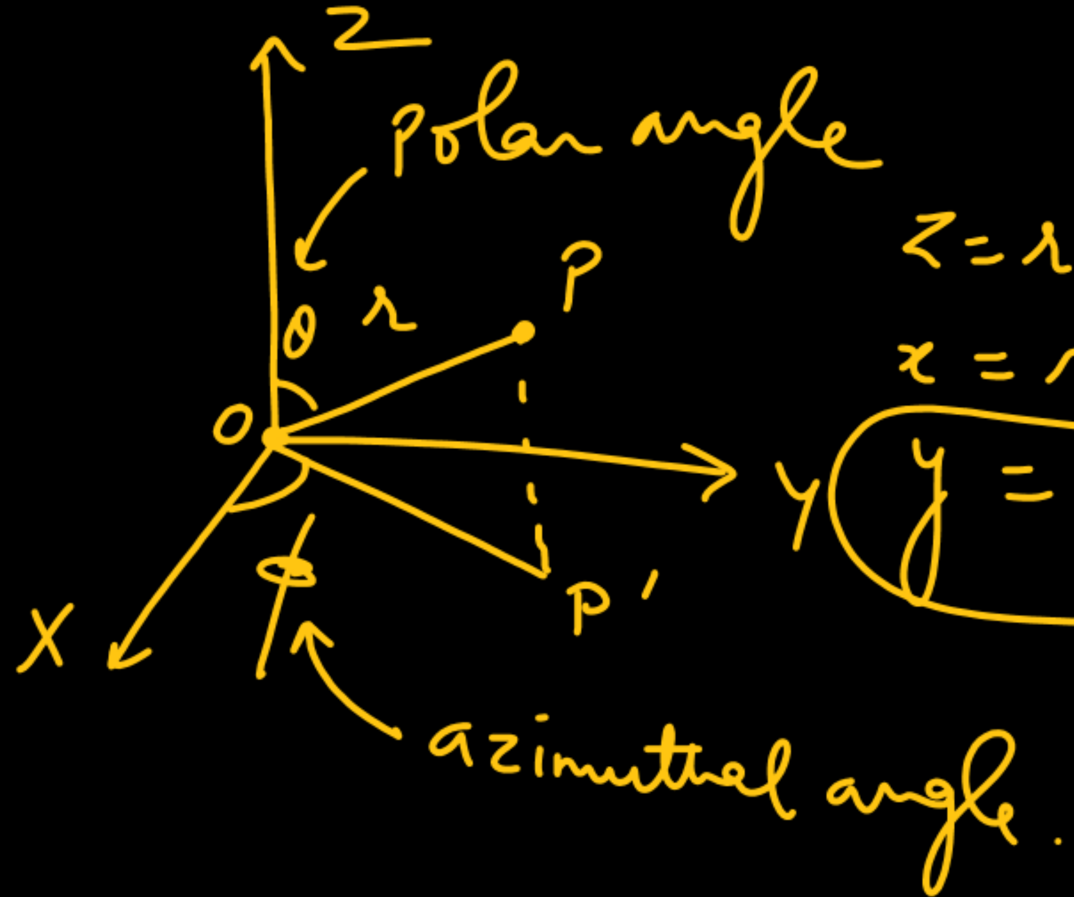
$\psi_r = K r^l$ (a polynomial of degree $(n-l-1)$) $\times e^{-r/K'}$

$(n-l-1)$ for $3p_y$ $= 3-1-1 = 1$

Sol^n : $3p_y$

$\psi_{(r,\theta,\phi)} = \psi_r \times \psi(\theta,\phi)$

$(\psi)_{3p_y} = \left\{K r^1 (K' - r)\, e^{-r/K''}\right\} \times K''' \times \sin\theta \sin\phi$



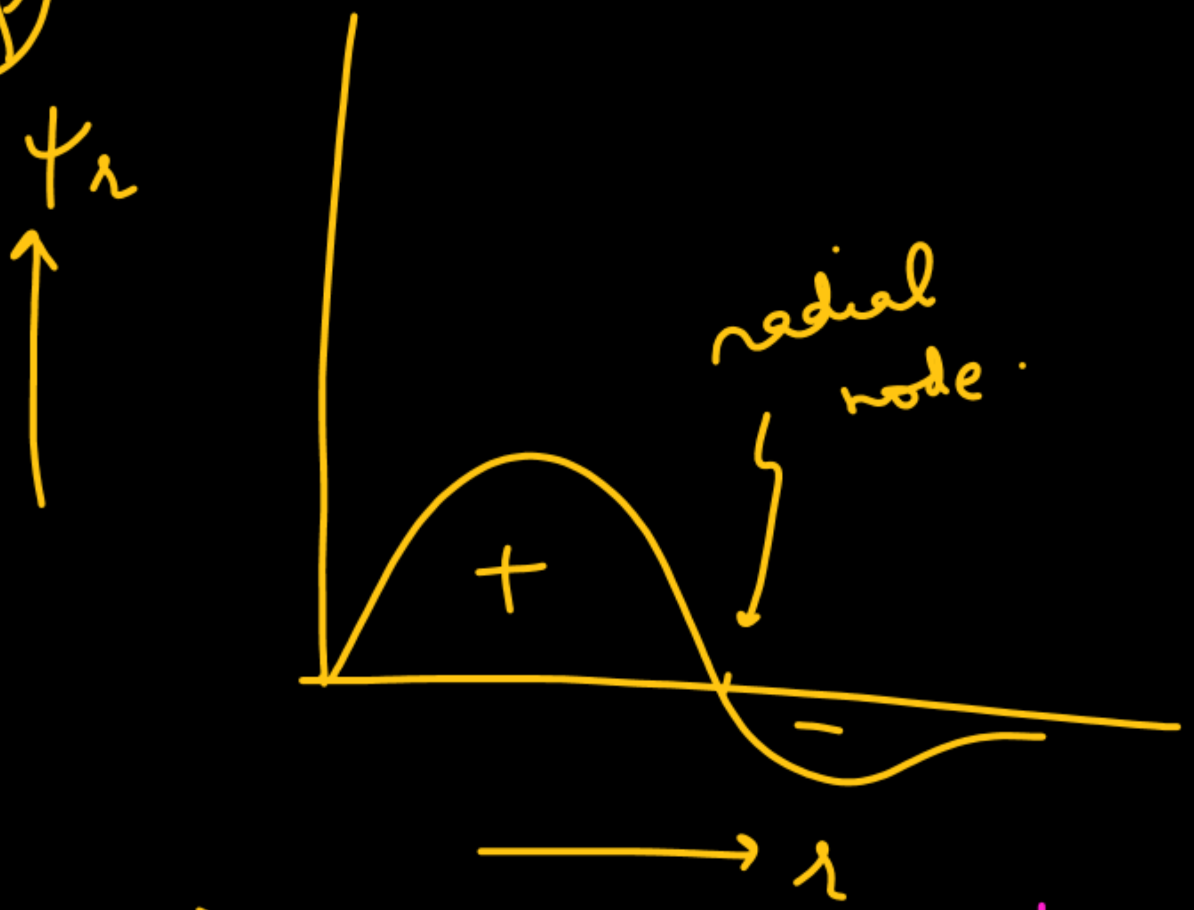
$z = r\cos\theta$

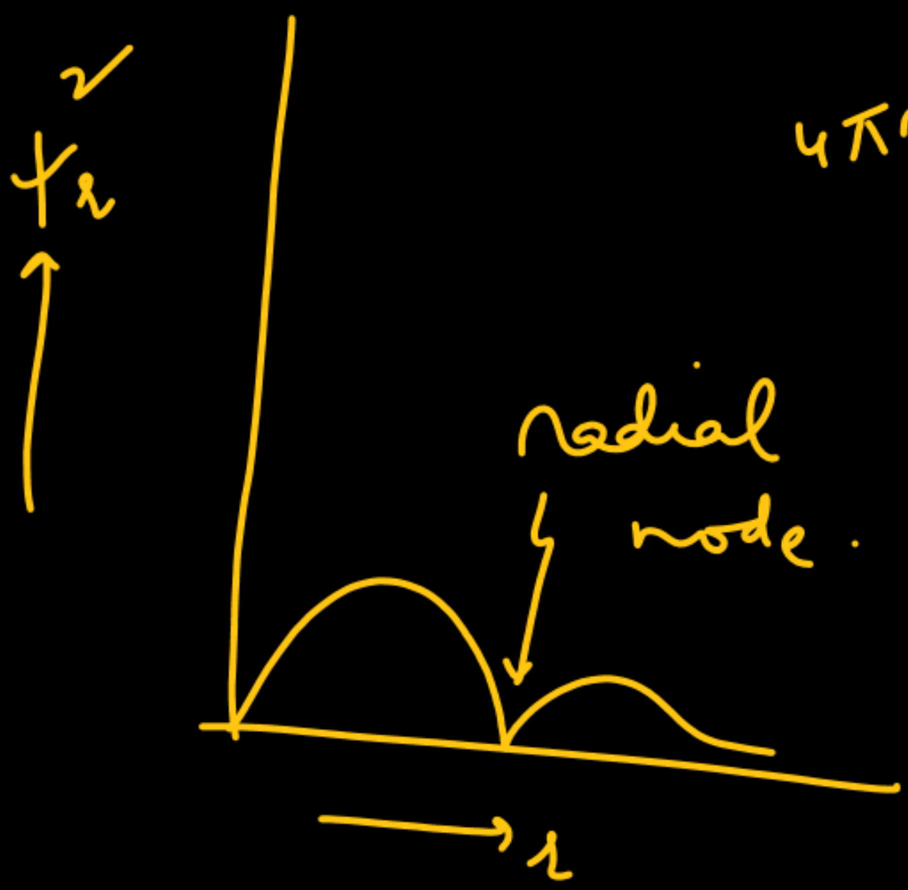
$x = r\sin\theta\cos\phi$

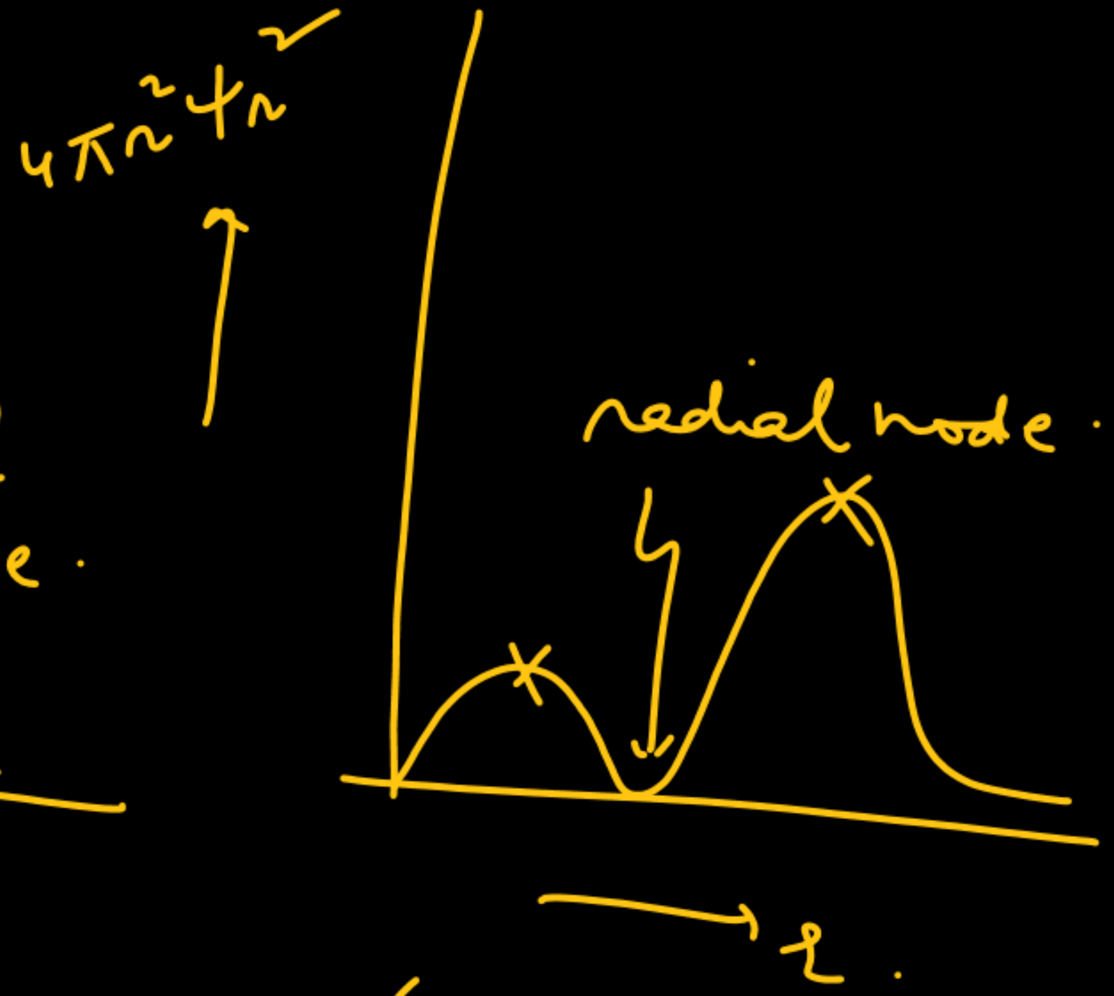
$y = r\sin\theta\sin\phi$

$(p_y) = K\sin\theta\sin\phi \rightarrow \psi(\theta,\phi)$

3p







(Two peaks) in 3py

$(n-l-1) = \text{Radial nodes}$
$= 3-1-1$
$= 1$

p_x p_y p_z

$-1 \; 0 \; +1$

$-1 \; 0 \; +1$

$0 \leftarrow$ 'm'

The hydrogen-like species Li^{2+} is in a sphericall symmetric state S_1 with one radial node. Upon absorbing light, the ion undergoes transition to a state S_2. The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom.

The state S_1 is

(A) 1s (B) 2s (C) 2p (D) 3s

Energy of the state S_1 in units of the hydrogen atom ground state energy is

(A) 0.75 (B) 1.50 (C) 2.25 (D) 4.50

The orbital angular momentum quantum number of the state S_2 is

(A) 0 (B) 1 (C) 2 (D) 3

Sol^n: Li^{2+} (spherically symmetric orbital : s orbital)

Radial nodes $= (n-l-1) = 1$

$n - 0 - 1 = 1$

$n = 2$

$\underbrace{S_1}_{ns \text{ i.e. } 2s} \xrightarrow{h\nu} S_2$

$\dfrac{-13.6 \times 3^2}{2^2} = \dfrac{9}{4} \times (E_1)_H = 2.25 \times (E_1)_H$

$n - l - 1 = 1$

$\Rightarrow n - l = 2$

$l = 1$

$S_2 = 3p$

$\dfrac{-13.6 \times 3^2}{n^2} = -13.6 \text{ eV}$

$n = 3$

The atomic masses of He and Ne are 4 and 20 amu, respectively. The value of the de Broglie wavelength of He gas at –73ºC is M times that of the de Broglie wavelength of Ne at 727ºC. The value of M is

$\boxed{M = 4}$ ✓

He ⟶ 4 amu
Ne ⟶ 20 amu.

$\lambda = \frac{h}{mv}$.

$u_{av} = \sqrt{\frac{8RT}{\pi M}}$.

$(\lambda_{He})_{200K} = M \times (\lambda_{Ne})_{1000K}$

$\frac{\cancel{h}}{4 \times \sqrt{\frac{\cancel{8R} \times \cancel{200}\,50}{\cancel{\pi} \times \cancel{4}}}} = M \times \frac{\cancel{h}}{20 \times \sqrt{\frac{\cancel{8R} \times \cancel{1000}\,5}{\cancel{\pi} \times \cancel{20}}}}$

$\frac{1}{4} = \frac{M}{20} \Rightarrow \underline{M = 5}$ Q..

Q: How many $\bar{e}$ in Cr atom in ground state have value $(l+m) = 0$.

Soln:

$_{24}Cr : 1s^2\, 2s^2\, 2p^6\, 3s^2\, 3p^6\, 3d^5,\, 4s^1$

$l = 0$

$m = 0$

$l = 1$

$m = -1, 0, +1$

2	2	2

2

$l = 2$

$m = -2, -1, 0, +1, +2$

↑	↑	↑	↑	↑

Ans: $2+2+2+2+2+1+1$

$= 12$ Ans